1. Project Overview

ShopShield AI is a lightweight, conversational safety tool that helps shoppers evaluate an unfamiliar store link before they pay. The user pastes a URL into a chat interface; the system checks it against a set of risk signals, converts those into a single **Trust Score (0–100)**, and uses an LLM to turn the findings into a short, plain-language explanation and recommendation.

GOALS

- Make link-safety checks instant for non-technical users.
- Reduce impulsive buying on unverified storefronts.
- Turn technical signals into one trustworthy number.
- Keep the whole tool buildable by one person.

THE PROBLEM IT SOLVES

Flash-sale and drop-shipping scam sites are cheap to spin up and often vanish within days. Shoppers have no quick way to check domain history, contact legitimacy, or pricing red flags — they rely on gut feeling. ShopShield AI replaces that with a structured, explainable check that takes seconds.

2. System Architecture

1 User Input
Paste a URL into the Streamlit chat UI. No sign-up, no extension.

2 URL Analysis
Checks domain age, SSL validity, contact info, and discount-keyword density.

3 Risk Scoring
Weighted, rule-based engine outputs a 0–100 Trust Score and risk level.

4 LLM Layer
Gemini/Ollama explains the pre-computed findings — it doesn't decide risk itself.

5 Output
Trust Score, plain-language reasons, and one clear safety recommendation.

3. Worked Example

User `www.xyz-super-sale.com`

Bot · Trust Score **25/100**

RISK LEVEL: HIGH

Reasons

- Domain name looks suspicious (urgency wording, not a brand name).
- No verified contact information found.
- Too many discount-related keywords on the landing page.

Recommendation: Avoid entering payment details. Verify the seller independently first.

4. Risk Scoring Logic

Scoring is rule-based, not LLM-based, so results stay consistent, explainable, and resistant to prompt manipulation.

Signal	What It Checks	Weight
Domain age	How recently the domain was registered (WHOIS)	20 pts
SSL / HTTPS	Whether the certificate is valid or missing	15 pts
Contact verification	Working support email, phone, or address	20 pts
Keyword patterns	Density of urgency/discount language	15 pts
Domain structure	Lookalike spelling, excess hyphens/subdomains	15 pts
External reputation	Cross-check vs. known scam/blocklist sources	15 pts

Score	Risk	Recommendation
80–100	Low	Safe to proceed with normal caution
50–79	Medium	Proceed carefully; verify seller first
0–49	High	Avoid entering payment or personal details

5. Stack, LLM Role & Roadmap

TECH STACK

- **UI:** Streamlit (Python-native chat)
- **URL parsing:** tldextract, requests
- **Domain intel:** python-whois
- **SSL:** ssl + socket (built-in)
- **Scoring:** plain Python function
- **LLM:** Gemini API or local Ollama
- **Cache:** SQLite

ROLE OF THE LLM

The LLM never decides risk itself — it only explains the structured score and signals it's given. This avoids hallucinated judgments and keeps output grounded.

Gemini: fast, free tier, needs internet.

Ollama: free, offline, no rate limits.

BUILD ROADMAP (SOLO)

- **Wk 1–2:** Streamlit UI + URL analysis + scoring.
- **Wk 2–3:** LLM integration + prompt tuning.
- **Wk 3–4:** Error handling + SQLite caching.
- **Wk 4+:** UI polish, deploy, gather feedback.

6. Key Challenges & Mitigations

Challenge	Issue	Mitigation
False sense of security	High scores may be over-trusted.	Always pair the score with specific reasons; label it a risk indicator, not a guarantee.
New legitimate domains	Brand-new honest stores may score poorly on age alone.	Weight domain age moderately; combine with contact & content checks before flagging High Risk.
LLM hallucination	Model could add unsupported claims.	Constrain the prompt to only explain the given structured signals — never invent new ones.
Cost / rate-limit creep	Frequent WHOIS/LLM calls add up.	Cache results per domain for ~24 hours in SQLite to avoid redundant re-analysis.

CLOSING NOTE

ShopShield AI is scoped to be buildable by one person in a few focused weeks. Its value comes from combining a few honest, verifiable signals with a clear explanation — so any shopper can make a faster, safer decision before they pay.